Alexander Massoumi

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Education

University of California, Irvine | Irvine, CA — Expected 2027
- B.S. Mechanical Engineering and B.S. Aerospace Engineering; Minor in Materials Science & Engineering | GPA: 3.76/4.00
- Tau Beta Pi · Sigma Gamma Tau (Aerospace Honor Society) · Certified SolidWorks Associate (CSWA)

Internship & Research Experience

Toyota Racing Development USA — Jun 2026 – Sep 2026
Mechanical Engineering Intern — *Costa Mesa, CA*
- Analyzed engine component failure data from SEM, confocal, and optical microscopy and Rockwell hardness tests to identify root causes, then documented findings in technical reports used in engineering design reviews.
- Restored an RDE-OES oil analysis spectrometer to calibrated service by recommissioning, troubleshooting, and standardizing it, enabling oil analysis to run in-house; also prepared metallographic specimens for microstructure review.

Advanced Casting Research Center (ACRC) — Jul 2026 – Present
Undergraduate Research Assistant — *Irvine, CA*
- In the Valdevit group, analyze metal cold spray deposition behavior across process parameter sets (gas temperature, pressure, standoff) to identify settings that improve deposit quality.
- Characterize rotating detonation engine (RDE) inner bodies before and after hot-fire tests to diagnose material and structural condition and support design iteration.

MD Anderson Cancer Center — Jun 2025 – Aug 2025
Physics & Engineering Research Intern — *Houston, TX*
- Built an automated Python pipeline that processes CT radiomic features, reduces them with LASSO and Sequential Backward Selection, and trains KNN and neural-network models to grade pancreatic cysts as high- or low-risk.

Engineering Teams

UCI Solar Racing (Zotsun Solar Car) — Aug 2025 – Present
Suspension & Steering Lead — *Irvine, CA*
- Lead a suspension and steering subteam through design iteration, validation, and fabrication of an electric solar car (Formula Sun Grand Prix), taking hardware from SolidWorks CAD to GD&T drawings, machined parts, assembly, and testing.

Flying Leatherneck Aviation Museum — Mar 2025 – Present
Fuselage Engineer, Curtiss JN-4 "Jenny" Replica
- Translated 2D engineering drawings into fabricated fuselage frame members by analyzing structures, building 3D CAD models, and machining and assembling parts.

Fixed-Wing Innovation Project (FWIP), UC Irvine — Jan 2025 – Present
Fuselage Engineer — *Irvine, CA*
- Designed and 3D-printed a lightweight fuselage structure for a fixed-wing UAV, iterating the design across flight-tested builds based on test results.

Projects

Autonomous Rover | *Arduino (C), Sensors & Actuators, 3D Printing* — Sep 2023 – Mar 2024
- Built and instrumented a line-tracking, can-retrieving rover, wrote Arduino (C) control code integrating sensors (magnetometer, limit switch) and actuators (solenoid, servos), and tested and iterated the system until it completed the task.

Transit Bus CAD Assembly & FEA | *SolidWorks, Finite Element Analysis, Motion Studies* — Sep 2024 – Dec 2024
- Modeled the UCI Anteater Express bus with working doors, wipers, wheels, and bike rack, and ran FEA to identify stress concentrations and maximum loading across structural sub-assemblies.

Skills

Data Analysis & Modeling: MATLAB, Python, test data analysis, machine learning, feature selection, technical documentation, validation findings and reports
Test & Validation: instrumentation, prototype test setup, instrument calibration, failure analysis, root cause analysis, SEM, confocal/optical microscopy, metallography, Rockwell hardness
CAD & Simulation: SolidWorks (CSWA), Finite Element Analysis (FEA), GD&T, 3D modeling, engineering drawings
Build & Controls: prototype builds, machine shop equipment, additive manufacturing (3D printing, cold spray), laser cutting, Arduino (C/C++), sensors and actuators, Git/GitHub

Leadership & Activities

- UC Irvine Anteater Express Operations Staff (Aug 2024 – Present): Safely operate Class B shuttles with 250+ passengers per shift.
- Engineering Student Council, Faculty Engagement (Oct 2023 – Present): Connect students with faculty research and industry.
- MotoGP Track Marshal, Circuit of the Americas (Mar 2025 – Present): Supported on-track incident response through SCCA.